

BORONG ZHANG

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ACADEMIC APPOINTMENTS

Georgia Institute of Technology

Visiting Assistant Professor, School of Mathematics

- Mentor: Wei Zhu

Atlanta, GA

08/2026—Present

EDUCATION

University of Wisconsin–Madison

Ph.D. in Mathematics

- Advisors: Qin Li and Leonardo Andrés Zepeda Núñez
- Honor: Mathematics Department Ascending Scholar Fellowship

Madison, WI

09/2020–06/2026

University of California, Berkeley

Bachelor of Arts in Applied Mathematics & Computer Science

- Research advisor: Rainer K. Sachs
- Honors: High Honors in Applied Mathematics; Distinction in General Scholarship; Dean's Honors List

Berkeley, CA

08/2016–06/2020

PUBLICATIONS

Zhang, B., Li, Q., & Di, Z. W. (2025). MAGPIE: Multilevel-Adaptive-Guided Solver for Ptychographic Phase Retrieval. *SIAM Journal on Scientific Computing*, accepted. <https://arxiv.org/abs/2504.10118>.

Zhang, B., Guerra, M., Li, Q., & Zepeda-Núñez, L. (2025). Back-Projection Diffusion: Solving the wideband inverse scattering problem with diffusion models. *Computer Methods in Applied Mechanics and Engineering*, 443, 118036. <https://doi.org/10.1016/j.cma.2025.118036>.

Zhang, B., Zepeda-Núñez, L., & Li, Q. (2024). Solving the wide-band inverse scattering problem via equivariant neural networks. *Journal of Computational and Applied Mathematics*, 451, 116050. <https://doi.org/10.1016/j.cam.2024.116050>.

Huang, E. G., Wang, R.-Y., Xie, L., Chang, P., Yao, G., **Zhang, B.**, Ham, D. W., Lin, Y., Blakely, E. A., & Sachs, R. K. (2020). Simulating galactic cosmic ray effects: Synergy modeling of murine tumor prevalence after exposure to two one-ion beams in rapid sequence. *Life Sciences in Space Research*, 25, 107–118. <https://doi.org/10.1016/j.lssr.2020.01.001>.

Preprints

Zhang, B., Deng, J., Jiang, Y., & Di, Z. W. (2025). Stochastic Multigrid Method for Blind Ptychographic Phase Retrieval. *arXiv*. <https://arxiv.org/abs/2511.01793>.

SELECTED TALKS

SIAM Conference on Mathematics of Data Science

11/2026

- Stochastic Multigrid Methods for Ptychographic Phase Retrieval

Salt Lake City, UT

SIAM Central States Section Annual Meeting

10/2025

- Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering

Fayetteville, AR

Institute for Mathematical and Statistical Innovation

06/2025

- Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering (Lightning Talk)

Chicago, IL

SIAM Conference on Applications of Dynamical Systems

05/2025

- Efficient Symmetry-Driven Diffusion Models for Wideband Inverse Scattering

Denver, CO

Copper Mountain Conference on Iterative and Multigrid Methods

04/2025

- Multigrid-based Stochastic Minimization for Ptychographic Phase Retrieval

Frisco, CO

PROFESSIONAL SERVICE

Co-organizer, minisymposium, SIAM Conference on Mathematics of Data Science

11/2026

- Model-Based and Data-Driven Methods for Inverse Problems and Imaging

Salt Lake City, UT

Reviewer, International Conference on Machine Learning

03/2026

RESEARCH EXPERIENCE

Givens Associate, MCS Division, Argonne National Laboratory	05/2025–08/2025
Supervisor: Dr. Zichao (Wendy) Di	Lemont, IL
- Proposed a stochastic multigrid method for blind ptychographic phase retrieval that employs a geometric-mean, phase-aligned joint object–probe update, with convergence guarantees. - Designed, implemented, and benchmarked the algorithm in Python (NumPy/PyTorch).	
Research Aide, MCS Division, Argonne National Laboratory	06/2024–09/2024
Supervisor: Dr. Zichao (Wendy) Di	Lemont, IL
- Proposed a stochastic multigrid, majorization–minimization framework for ptychographic phase retrieval. - Delivered a comprehensive theoretical analysis of convergence. - Designed, implemented, and benchmarked the algorithm in Python.	

TEACHING EXPERIENCE

Georgia Institute of Technology	Atlanta, GA
Math 2550: Introduction to Multivariable Calculus, Instructor	Fall 2026
University of Wisconsin–Madison	Madison, WI
Math 718: Randomized Linear Algebra and Applications, TA	Spring 2026
Math 340: Elementary Matrix and Linear Algebra, TA	Fall 2025
Math 211: Calculus, Head TA	Spring 2025
Math 112: College Algebra, Instructor	Fall 2023, Spring 2024, Fall 2024
Math 234: Calculus: Functions of Several Variables, TA	Spring 2023
Math 211: Calculus, TA	Fall 2022
Math 521: Analysis I, TA	Summer 2022
Math 222: Calculus and Analytic Geometry II, TA	Spring 2021
Math 221: Calculus and Analytic Geometry I, TA	Fall 2020

MENTORING & OUTREACH

University of Wisconsin–Madison	Madison, WI
Graduate Peer Mentor	Fall 2025
Mentor, Directed Reading Program	Fall 2024, Spring 2025